Secure, Policy-Driven Student Placement Management System

1. Problem Definition

The Placement Cell of NFSU requires a centralized, secure, and policy-driven web application to maintain course-wise student placement records and manage recruitment activity in accordance with the Student Placement Guidelines for the 2026–27 placement session. The system must maintain current student profiles, recruitment participation, offers, internship and full-time employment (FTE) details, pre-placement offers (PPOs), package information, and placement eligibility/status.

The application should automate policy checks, reduce manual data-entry errors, flag possible policy conflicts, prevent unauthorized changes, and generate reliable placement statistics. Because the system processes personal contact information and compensation details, security and privacy must be built into its design.

2. Programmes Covered

- M.Tech. Computer Science (M.Tech. CS)
- M.Tech. Artificial Intelligence and Data Science (M.Tech. AIDS)
- B.Tech.–M.Tech. Computer Science and Engineering (CS)
- M.Sc. Computer Science (M.Sc. CS)
- M.Sc. Digital Forensics and Information Security (M.Sc. DFIS)

3. Core Functional Requirements

3.1 Student Database Management

Maintain a searchable, up-to-date student database. Support authorized creation, viewing, updating, archiving, and controlled import/export of records. Preserve placement history rather than overwriting previous offer events.

DATA FIELD	REQUIREMENT
Placement ID	Unique identifier; reject duplicates.
Student name	Full name as maintained by the institution.
Mobile number	Registered contact number; access restricted by role.
Email ID	Institutional or registered email address.
Course and batch	One of the five programmes listed above and the relevant academic batch.
Registration status	Registered, deregistered, or not registered.
Eligibility status	Eligible, ineligible, or pending verification.
Placement status	Unplaced, internship only, FTE only, or internship + PPO/FTE.
Company and offer type	Recruiting organization and I, F, or I+F classification.
Internship stipend	Amount and payment period, where applicable.
FTE package / CTC	Annual compensation in LPA, where applicable.
PPO status	Not applicable, pending, offered, accepted, or declined.

Offer acceptance	Pending, accepted, declined, or expired.
Joining status	Pending, joined, or not joined.
Offer date	Date on which the offer was issued.
Remarks / audit reference	Authorized administrative notes and links to related audit records.

3.2 Placement Policy Automation

Implement configurable rules based on the attached NFSU Student Placement Guidelines 2026–27. The application should explain why a student is allowed, blocked, or referred for manual review.

Company category	**Package / CTC**	**Switching rule in the policy**
A+	≥ ₹16 LPA	No switching policy; students selected in A+ are out of the placement process.
A	₹10–16 LPA	Switching allowed only to A+ category companies.
B	₹6–10 LPA	Further placement allowed where the new package is at least 1.5 times the existing package.
C	Below ₹6 LPA	Eligible for higher-category opportunities as described in the policy.

Offer-type restrictions to implement:

- Internship only (I): After selection for an internship, the student cannot apply for another internship-only role, but may apply for FTE or I+F roles, subject to other policy rules.
- FTE only (F): After selection for FTE, the student cannot apply for another FTE-only role, but may apply for an internship role, subject to other policy rules.
- Internship + PPO/FTE (I+F): After selection, the student cannot participate in other I, F, or I+F roles at another organization.

Additional placement checks:

- Verify placement registration and company-specific eligibility before participation.
- Record attendance at pre-placement talks, presentations, tests, interviews, and other recruitment activities.
- Flag absenteeism for possible exclusion from the next three drives, while supporting policy-defined emergency exceptions and authorized review.
- Track offer acceptance deadlines: 24 hours or the company's stated timeline, whichever is earlier.
- Record off-campus offers, deregistration, joining updates, and relevant NOC status where applicable.
- Flag suspected policy violations for review; final disciplinary decisions and exceptions remain with the Placement Cell/competent authority.

3.3 Dashboard and Reporting

- Total registered, eligible, ineligible, placed, and unplaced students.
- Course-wise and batch-wise placement outcomes.
- Counts for internship only, FTE only, and internship + PPO/FTE.

- Company-wise offers, categories, and compensation packages.
- Highest, average, and median FTE packages, with stipend and annual CTC reported separately.
- Students eligible for subsequent drives under switching rules.
- Pending offer acceptance, joining, and policy-compliance cases.

Reports must distinguish internship stipends from annual FTE CTC and must count unique placed students separately from the total number of offers received.

4. Cybersecurity and Privacy Requirements

SECURITY AREA	EXPECTED CONTROL
Authentication	Secure login, appropriate password hashing, and optional multi-factor authentication.
Role-based access control	Separate permissions for Placement Administrator, Placement Officer, Course Coordinator, and Company / Recruiters.
Confidentiality	Encrypt data in transit and at rest; restrict access to contact and compensation details.
Input and query security	Server-side validation, parameterized database queries, and protection against SQL injection and cross-site scripting.
Audit logging	Record sensitive access and changes, including actor, timestamp, action, and relevant before/after values where appropriate.
Integrity	Unique Placement IDs, duplicate detection, validation rules, and tamper-evident audit records.
Session security	Secure session handling, logout, timeout, and protection against session hijacking.
Backup and recovery	Encrypted backups and a tested restoration procedure.
Export security	Restrict, protect, and audit CSV/Excel exports containing personal or salary data.
Privacy and retention	Collect only necessary data and define authorized retention and deletion procedures.

Security challenge: A student must not be able to change their own placement status or package, bypass switching restrictions, or access another student's confidential information by manipulating requests or attempting to access unauthorized records.

5. Suggested System Workflow

1. Student registration and data import	Validate Placement ID, course, contact details, and registration status.
2. Eligibility verification	Check registration, academic eligibility, company-specific criteria, and approved exceptions.
3. Recruitment and offer entry	Record company, category, offer type, package, offer status, and dates.

4. Policy rule engine	Evaluate switching rules and flag conflicts before the next drive.
5. Secure dashboard and reports	Generate course-wise statistics, audit trails, and authorized reports.

6. Expected Deliverables

1. Working prototype of a student database with create, read, update, and controlled delete/archive operations and course-wise filtering.
2. Secure login, role-based access controls, and an administrative dashboard.
3. Policy rule engine covering offer categories, offer types, and switching eligibility.
4. Recruitment and offer-management module supporting I, F, and I+F.
5. Course-wise and university-level placement reports.
6. Audit logs and validation against unauthorized modifications.
7. Test cases demonstrating policy enforcement and resistance to common web application attacks.
8. Technical documentation covering database schema, architecture, security controls, testing, and limitations.

7. Expected Outcome

Develop a secure, policy-driven placement management prototype that enables NFSU's Placement Cell to maintain accurate course-wise student records, apply placement guidelines consistently, protect sensitive information, identify policy conflicts, and produce reliable placement statistics across all five programmes.